
Federated Learning for Privacy-Preserving Distributed Space Weather Monitoring in Smart Grid Protection

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September 2026

Abstract — The vulnerability of modern power grids to geomagnetically induced currents (GICs), triggered by M- and X-class solar flares, presents a critical and recurring threat to bulk electric infrastructure; the 1989 Québec system collapse remains the canonical demonstration of how a single geomagnetic storm can disable an entire provincial grid. Effective early-warning systems require predictive models trained on high-fidelity solar observations spanning multiple geographic regions, yet nationally sensitive space-weather data collected by agencies such as NASA, ESA, JAXA, ISRO, KASI, and the Australian Bureau of Meteorology cannot be centralised because of data-sovereignty constraints and inter-agency data-sharing agreements. This paper presents, to the best of our knowledge, the first application of federated learning (FL) to solar flare prediction for smart-grid protection, explicitly addressing this privacy barrier. The proposed framework partitions the SWAN-SF benchmark dataset into six geographically distributed regional-observatory clients using Dirichlet sampling, inducing realistic non-IID label skew. Each client trains a local neural model on 144 statistical features derived from 24 photospheric magnetic-field parameters, and transmits only weight updates—never raw satellite measurements—to an aggregation server. All results are produced under a frozen evaluation protocol: an immutable train/validation/test contract, thresholds and calibration selected on validation only, a runtime leakage audit, and single-pass test evaluation. On the held-out test set (331,185 windows at the natural 1.88% flare prevalence), FedProx attains a ROC-AUC of 0.954 (97.7% of the centralised XGBoost reference’s 0.977) and a PR-AUC of 0.307, exceeding the pooled centralised MLP of identical architecture (0.888 / 0.153)—so federation costs no ranking skill relative to the same hypothesis class. Plain FedAvg diverges under the same partition (validation AUC 0.99 $\rightarrow$ 0.05 after round 20); the proximal term is the load-bearing design choice, beating FedAvg on all five seeds (Wilcoxon $p = 0.031$) with $2.7\times$ narrower metric variance. A 24-cell ablation grid isolates the contribution of each component (focal loss, DA-FL aggregation, SMOTE, proximal term); SMOTE is inert on the pre-balanced benchmark and we report that null explicitly. Cross-model SHAP analysis shows the federated model’s decisions concentrate on the same physical drivers as the centralised surrogate—current helicity, the R-value flux-emergence proxy, and Lorentz-force terms. We further document a threshold-transfer failure under the $26\times$ validation-to-test prevalence shift and report prevalence-robust recall-at-FPR operating points (FedProx: 0.51 recall at 2% FPR). These results demonstrate that distributed space-weather agencies can jointly train a usable flare-prediction model without pooling raw observations, establishing federated machine learning as a practical mechanism for smart-grid resilience against geomagnetic threats—with data-locality, not cryptographic privacy, as the verified property, and with the full flare-to-GIC chain remaining future work.

Keywords: federated learning; solar flare prediction; smart grid; geomagnetically induced currents; FedProx; space weather; SWAN-SF; non-IID data; class imbalance; data privacy

Source code and all experimental artefacts: <https://github.com/Daveshvats/Federated-Learning-for-Distributed-Space-Weather-Monitoring>

1 Introduction

Modern society has quietly built its electrical backbone in the path of a cosmic hazard. Geomagnetically induced currents (GICs) are quasi-direct currents that flow in high-voltage transmission lines when a geomagnetic storm—itsself driven by the arrival of a coronal mass ejection (CME) associated with a major solar flare—distorts the Earth’s magnetosphere. The time-varying geomagnetic field induces geo-electric fields in the conducting crust, and these fields drive GICs through grounded transformer neutrals. Once inside a high-voltage transformer, even a modest GIC biases the alternating flux into half-cycle saturation, producing harmonics, reactive-power absorption, and localized heating that can culminate in irreversible thermal damage [1, 2]. The canonical demonstration of this failure mode remains the 13 March 1989 storm, which collapsed the Hydro-Québec grid in ninety seconds and left six million customers without power for nine hours [1, 2]. The July 2012 event, an eruptive flare/CME of Carrington-class magnitude that missed Earth’s orbital position by roughly nine days, is a stark reminder that the next exposure is a matter of when, not if [3].

The first line of defence against such events is forecasting. Grid operators who receive an early warning that an M- or X-class flare has erupted, or is likely to erupt, in a magnetically complex active region can pre-emptively derate transformers, adjust reserve margins, postpone switching operations, and prepare load-shedding plans in the window between the flare observation and the CME’s arrival one to three days later. Machine-learning flare-prediction models built on photospheric vector-magnetogram summaries—the SHARP parameter suite [4] and its descendants—have demonstrated statistically skilful classification of flaring versus non-flaring active regions [5, 6]. The predictive signal lives in quantities such as total unsigned current helicity, magnetic free energy, and flux-emergence proxies, all of which are computed routinely by solar observatories on every continent and in orbit.

There is, however, a structural obstacle to training the strongest possible model: the data cannot be pooled. The highest-fidelity flare-forecasting observations are produced and curated by nationally mandated agencies—NASA and NOAA in the Americas, ESA in Europe, JAXA in the Asia-Pacific region, ISRO in South Asia, KASI in East Asia, and the Australian Bureau of Meteorology (BoM) in Oceania. Raw magnetogram-derived time series are treated as nationally sensitive assets in many jurisdictions, and inter-agency data-sharing agreements historically favour metadata exchange over bulk data transfer. Consequently, each agency trains (or could train) models on its own regional shard, and the global community inherits the fragmentation of six models each seeing a biased slice of the active-region population. A genuinely global model, trained on all regions’ data without any region surrendering that data, has been out of reach.

Federated learning (FL) [7, 8] is precisely the mechanism for this setting. In cross-silo FL, each data-holder trains a shared model architecture locally and transmits only parameter updates to an aggregation server, which averages them into a new global model. Raw measurements never leave the custodian’s infrastructure. FL has already proven transformative in medicine and digital health, where patient-privacy constraints are structurally identical to the data-sovereignty constraints of space-weather agencies [9]. What has not existed, prior to the present work, is a systematic treatment of FL for solar flare prediction—the solar-physics community’s benchmark workflows [5, 6, 10] are all centrally pooled, and the FL literature contains, to the best of our knowledge, no application to photospheric flare forecasting.

This paper closes that gap. We present a complete, reproducible federated framework—implemented, versioned, and released as open source—that simulates six geographically distributed regional observatories, each holding a non-IID shard of the SWAN-SF benchmark [10] in its cleaned and rebalanced form, and evaluates whether federation can approach the centralised upper bound without any data movement. The specific contributions are:

1. **A problem formulation and system architecture** that maps smart-grid GIC early warning onto a cross-silo federated classification problem with a safety-critical, recall-weighted objective, and that abstracts the data-sovereignty constraint of national space-weather agencies into a concrete, simulable protocol (Section 3 and Section 4).
2. **A non-IID federation protocol for SWAN-SF** that combines Dirichlet label-skew partitioning across six regional clients, per-client SMOTE oversampling, a server-aware Fed-Focal loss [11] for extreme class imbalance, and distribution-aware aggregation weighting, on top of both FedAvg [7] and FedProx [12] training loops (Section 4).
3. **A frozen, leakage-audited evaluation protocol:** an immutable train/validation/test contract, threshold and calibration selection on validation only, single-pass test evaluation, a centralised-MLP architecture control, five-seed statistics with confidence intervals and paired significance tests, and a 24-cell component-ablation grid (Section 5).
4. **An empirical privacy-cost quantification.** On the held-out test partitions (331,185 windows, 1.88% prevalence), FedProx achieves a ROC-AUC of 0.954—97.7% of the centralised

XGBoost reference (0.977) and *above* the pooled centralised MLP of identical architecture (0.888)—while plain FedAvg diverges during training (validation AUC $0.99 \rightarrow 0.05$); Fed-Prox beats FedAvg on all five seeds (Wilcoxon $p = 0.031$) (Section 6).

5. **A cross-model interpretability audit.** SHAP analysis of both the centralised XGBoost reference and the federated global model, with physical-parameter-group aggregation, shows both concentrate decisions on current helicity, the Schrijver flux-emergence proxy [13], and Lorentz-force components—consistent with established flare physics (Section 6).

In addition to these scientific contributions, the released framework embodies a documented engineering narrative: a six-version stability iteration (v2.1–v2.6) in which Dirichlet concentration, focal-loss alpha clamping, optimiser choice for variance-reduced FL, and dataset auditing were each adjusted in response to diagnosed failure modes. We summarise this evolution in Appendix A because it constitutes, in our view, transferable know-how for practitioners attempting the first FL deployment in any physical-science domain.

The remainder of the paper is organised as follows. Section 2 reviews flare prediction and federated learning. Section 3 formalises the problem. Section 4 details the system architecture and methodology. Section 5 describes the experimental setup, Section 6 presents and analyses results, and Section 7 discusses operational integration and engineering lessons. Section 8 states limitations and future work, and Section 9 concludes.

2 Background and Related Work

2.1 Solar flare prediction from photospheric magnetograms

The physical premise of data-driven flare forecasting is that the photospheric magnetic field of an active region encodes the stress budget that a future flare will release. Schrijver [13] demonstrated that a single scalar—the R-value, the unsigned flux within high-gradient polarity-separation corridors—separates major-flaring regions from non-flaring ones with striking reliability, and this proxy remains among the strongest single flare predictors identified to date. Bobra and Couvidat [4] operationalised the machine-learning view of the problem, showing that a support-vector machine trained on 25 SHARP parameters derived from SDO/HMI vector magnetograms can forecast M-class flares with useful skill, and that feature selection matters as much as model choice. Subsequent community-wide blind tests, most notably the multi-method comparison of Leka et al. [5], established two durable findings: (i) no single method dominates across all event definitions and skill metrics, and (ii) the effective ceiling on skill is set by the quality and homogeneity of the training data rather than by classifier complexity.

The European FLARECAST project [6] pushed the operational agenda further, combining multiple forecasting methods and data sources in an open framework, and it explicitly confronted the data-heterogeneity problem that motivates the present paper. On the data side, the SWAN-SF benchmark of Angryk et al. [10] released a multivariate time-series dataset of roughly 8,000 active-region tracks with 24 physical parameters per timestep and flare-history labels, specifically designed to make flare prediction reproducible and comparable; it has since become the community’s reference benchmark, and a family of cleaned and rebalanced variants (random undersampling, Tomek-link cleaning, and TimeGAN-based synthetic augmentation [14]) has been curated to make the benchmark tractable for supervised learning. Recent deep-learning work on SWAN-SF continues to raise the centralised ceiling: contrastive pretraining and multifaceted preprocessing [15] and LSTM-style temporal architectures [16] exploit the sequential structure of the 60-minute observation windows. All of these works, without exception, assume that the full training corpus is available in one place.

2.2 Federated learning under statistical heterogeneity

Federated learning was introduced by McMahan et al. [7] in the FedAvg formulation: clients perform several epochs of local SGD on private data, and a server aggregates the resulting weights by sample-size-weighted averaging. The defining difficulty, identified almost immediately, is statistical heterogeneity: when client data distributions differ—the non-IID setting—local optima diverge from the global optimum, and naive averaging can oscillate or diverge [17]. FedProx [12] addresses this by adding a proximal term $(\mu/2) \|\mathbf{w} - \mathbf{w}^{(t)}\|^2$ to each client’s local objective that anchors local training to the current global model, tolerating partial work and heterogeneous local computation. SCAFFOLD [18] attacks the same problem with control variates that correct client drift using variance reduction. The comprehensive survey of Kairouz et al. [8] situates these algorithms within the broader open-problem landscape, including robust aggregation, personalisation, and privacy. Cross-silo FL—few, reliable, institutionally owned clients—is the deployment regime relevant to space-weather agencies, and it contrasts with the cross-device regime in its stability requirements and long-lived participants.

2.3 Class imbalance in federated learning

Flare prediction inherits an extreme class-imbalance regime: in SWAN-SF’s natural test distribution, roughly 1.9% of windows precede M/X-class flares, while the rebalanced training partitions sit near parity. The focal loss of Lin et al. [19], originally developed for dense object detection, down-weights easy examples with a $(1 - p_t)^\gamma$ modulating factor and concentrates gradient on hard positives; it has become the default imbalance workhorse. Sarkar et al. [11] adapted focal loss specifically to FL in the Fed-Focal formulation, in which the class-weighting parameter is adjusted using information available at the server about the global class distribution, so that clients with different local positivity rates do not implicitly reweight the global objective. In parallel, synthetic oversampling remains a standard pre-processing response: SMOTE [20] interpolates minority-class neighbours, and mixup [21] creates convex combinations of training pairs. Under federation, rebalancing must be applied locally per client, since the pooled distribution is precisely what no participant can inspect.

2.4 Positioning of this work

Two research threads—physics-grounded flare prediction and heterogeneity-robust federated learning—have developed largely independently, and we are not aware of any prior work that joins them. Searches of the FL literature reveal applications spanning digital health [9], finance, and IoT sensing, but none addressing space-weather forecasting or solar flare prediction; conversely, the flare-forecasting literature, including its most recent deep-learning incarnations [15, 16], is entirely centralised. The present work is, to the best of our knowledge, the first federated treatment of solar flare prediction, the first FL framework evaluated on the SWAN-SF benchmark, and the first to quantify the privacy cost of federation—the gap between federated and centralised skill—specifically for a safety-critical smart-grid application. In doing so it also contributes a rarely documented artifact: an honest engineering log of the failure modes encountered when FL meets a severely imbalanced physical-science benchmark (Appendix A).

3 Problem Formulation

3.1 Federated risk minimisation

Let K denote the number of participating agencies (clients), each holding a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ of active-region observation windows $x_i \in \mathbb{R}^d$ and binary labels $y_i \in \{0, 1\}$

indicating whether an M- or X-class flare occurs within the forecasting horizon. Let $\mathcal{D} = \bigcup_{k=1}^K \mathcal{D}_k$ with $n = \sum_k n_k$. The goal of standard centralised learning is to solve

$$\min_{\mathbf{w} \in \mathbb{R}^p} F(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \ell(f_{\mathbf{w}}(x_i), y_i), \quad (1)$$

where $f_{\mathbf{w}}$ is the prediction model, ℓ a per-sample loss, and p the number of parameters. In the federated setting this objective is decomposed as a weighted sum of local empirical risks

$$\min_{\mathbf{w}} F(\mathbf{w}) = \sum_{k=1}^K \rho_k F_k(\mathbf{w}), \quad F_k(\mathbf{w}) = \frac{1}{n_k} \sum_{(x_i, y_i) \in \mathcal{D}_k} \ell(f_{\mathbf{w}}(x_i), y_i), \quad (2)$$

with aggregation weights $\rho_k \geq 0$, $\sum_k \rho_k = 1$. The vanilla FedAvg choice is $\rho_k = n_k/n$; Section 4 describes the distribution-aware weights used in this work.

3.2 The privacy constraint

The operational constraint that distinguishes cross-silo FL from distributed optimisation is not an algorithmic preference but an institutional invariant: for every round t and client k , the server may observe only the weight increment $\Delta \mathbf{w}_k^{(t)} = \mathbf{w}_k^{(t+1)} - \mathbf{w}_k^{(t)}$ (or equivalently the locally trained weights), never x_i or y_i for any sample in $\mathcal{D}_k$. Formally, the communication transcript

$$\mathcal{T} = \{\Delta \mathbf{w}_k^{(t)}\}_{k=1, \dots, K; t=1, \dots, T} \quad (3)$$

is the only quantity that crosses the agency boundary. This matches the data-sovereignty position of national space-weather agencies, for which raw magnetogram time series are nationally held assets, and it is the property that the released framework enforces by construction: the client-side pipeline never serialises raw shards.

3.3 Statistical heterogeneity

Because different agencies monitor different populations of active regions—different longitudes, hemispheres, and instrument footprints—the local class priors $p_k = \Pr(y = 1 \mid k)$ differ from the pooled prior $\bar{p}$. We simulate this label skew by partitioning the pooled training data among K clients with a symmetric Dirichlet distribution of concentration α (Section 4), so that the realised local priors span a wide range around $\bar{p}$. The consequence for optimisation is well documented [12, 17]: local objectives F_k become misaligned, and plain averaging of locally optimised weights introduces drift that grows with the number of local epochs. FedProx counteracts this drift by locally solving

$$\min_{\mathbf{w}} F_k(\mathbf{w}) + \frac{\mu}{2} \|\mathbf{w} - \mathbf{w}^{(t)}\|^2, \quad (4)$$

which keeps client updates in a trust region around the global model. The strength of this anchor, μ , is a central hyperparameter of this study.

3.4 A safety-critical, recall-weighted objective

Grid protection inverts the usual classification cost structure. A false-positive flare alert costs a operator a conservative derating or an unnecessary operational posture; a false negative can cost a fleet of high-voltage transformers. Following operational forecasting practice, we therefore evaluate with the F_β measure

$$F_\beta = \frac{(1 + \beta^2) P R}{\beta^2 P + R}, \quad \beta = 2, \quad (5)$$

where P and R are precision and recall; $\beta = 2$ weights recall twice as heavily as precision. Because the deployed decision rule is a threshold τ on the predicted flare probability, the framework treats τ as a free, per-model parameter selected on held-out validation evidence to maximise Eq. (5), rather than fixing $\tau = 0.5$ by convention. Section 6 shows this choice is not cosmetic: at the conventional threshold, every federated model trained on rebalanced local data is operationally unusable, and threshold optimisation is what converts a well-ranked model into a usable detector.

4 System Architecture and Methodology

4.1 Architecture overview

Figure 1 shows the complete system. Six regional observatory clients—stand-ins for NASA/NOAA, ESA (PROBA-2), JAXA, ISRO, KASI, and BoM—each hold a private, non-IID shard of the benchmark. In every communication round, each selected client trains a local copy of the shared model on its shard (with local rebalancing and an imbalance-robust loss, described below), and returns only its weight increment to the aggregation server. The server forms the next global model with distribution-aware weighted averaging, broadcasts it to all clients, and the cycle repeats for $T = 50$ rounds. Downstream of federation, the global model is calibrated for operations: its flare probabilities pass through the F_β -optimal threshold and become GIC early-warning decisions, which plug into existing SCADA workflows for grid management. The division of labour is deliberate: everything that touches raw magnetogram data (loading, rebalancing, local training) executes inside the client boundary; everything the server touches is a weight tensor.

4.2 Data: the SWAN-SF benchmark and its cleaned variant

All experiments use the SWAN-SF benchmark [10] (doi:10.7910/DVN/EBCFKM), a multivariate time-series dataset of active-region observation windows drawn from SDO/HMI vector magnetograms, with 24 physically motivated parameters per timestep sampled over 60-minute sliding windows and labels indicating the occurrence of $\geq M$ -class flares within the prediction horizon. The 24 parameters summarise magnetic-field complexity and include, among others: total unsigned current helicity (TOTUSJH), total photospheric magnetic free energy (TOTPOT), total unsigned vertical current (TOTUSJZ), total unsigned flux (USFLUX), the sum of absolute net-current-per-polarity (SAVNCPP), absolute net current helicity (ABSNJZH), mean shear angle and the fraction of high-shear area (MEANSHR, SHRGT45), the Schrijver flux-emergence proxy (R-value), and the three components of the total Lorentz force (TOTFX, TOTFY, TOTFZ) [4, 13].

We ingest the benchmark through its community-maintained cleaned variant¹, combining all five data partitions. The training partitions are class-balanced by this preprocessing (pooled positive rate $\bar{p} \approx 0.489$), while the untouched test partitions retain the natural, severely imbalanced prevalence of approximately 1.9% flare windows across roughly 3.3×10^5 held-out samples. This train–test contrast is not an artefact we introduce; it is inherited from the benchmark’s design philosophy, and it has a consequence that shapes the entire evaluation: models trained at $\sim 49\%$ prior must detect events at a $\sim 2\%$ prior, so absolute probability calibration degrades and decision thresholds must be re-optimised post hoc (Section 4.8).

¹Cleaned SWAN-SF Dataset, <https://github.com/samresume/Cleaned-SWANSF-Dataset>, which applies per-partition preprocessing: random undersampling with Tomek links and TimeGAN-based synthetic augmentation for the training partitions [14], LSBZM normalisation, and FPC-KNN imputation.

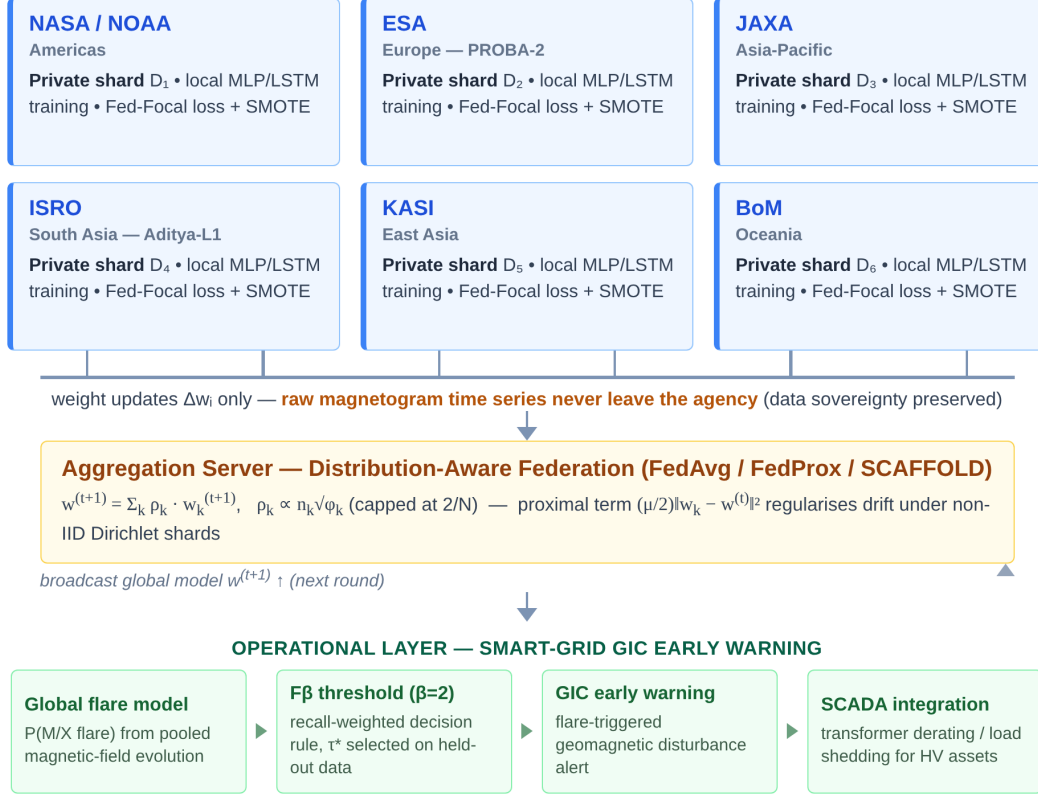


Figure 1: Federated space-weather monitoring architecture for smart-grid protection. Six regional observatory clients hold private non-IID shards of the benchmark; only weight updates $\Delta \mathbf{w}_k$ cross the agency boundary. The aggregation server applies FedAvg/FedProx with distribution-aware weighting ($\rho_k \propto n_k \sqrt{\phi_k}$, capped at $2/K$). The global model feeds an F_β -optimal decision layer for GIC early warning and SCADA integration.

4.3 Client-side feature representation

The framework supports two representations of the same 60-timestep observation windows. In the MLP path used for the reported run, each three-dimensional window $X \in \mathbb{R}^{60 \times 24}$ is reduced to a 144-dimensional statistical vector

$$\tilde{x} = [\mu, \sigma, \max, \min, \delta, s] \in \mathbb{R}^{144}, \quad (6)$$

where each block is applied per base feature: μ and σ are the temporal mean and standard deviation, $\max$ and $\min$ the window extrema, $\delta = X_T - X_1$ the net trend, and s the least-squares slope against normalised time. This six-statistic extraction preserves substantially more temporal information than a temporal mean (peaks, volatility, and rate of change are precisely the flare-relevant quantities) while remaining compatible with weight-averaged federation and with the centralised baselines. In the LSTM path, the raw $\mathbb{R}^{60 \times 24}$ tensor is consumed directly by a recurrent model; this path is fully implemented in the released code but the committed artefacts analysed here correspond to the MLP execution, and we flag the LSTM evaluation as future work (Section 8).

4.4 Non-IID partitioning into regional observatories

To emulate geographic heterogeneity, the pooled training set is partitioned by label-aware Dirichlet sampling: for each class c , a proportion vector is drawn from $\text{Dirichlet}(\alpha, \dots, \alpha) \in \mathbb{R}^K$ and

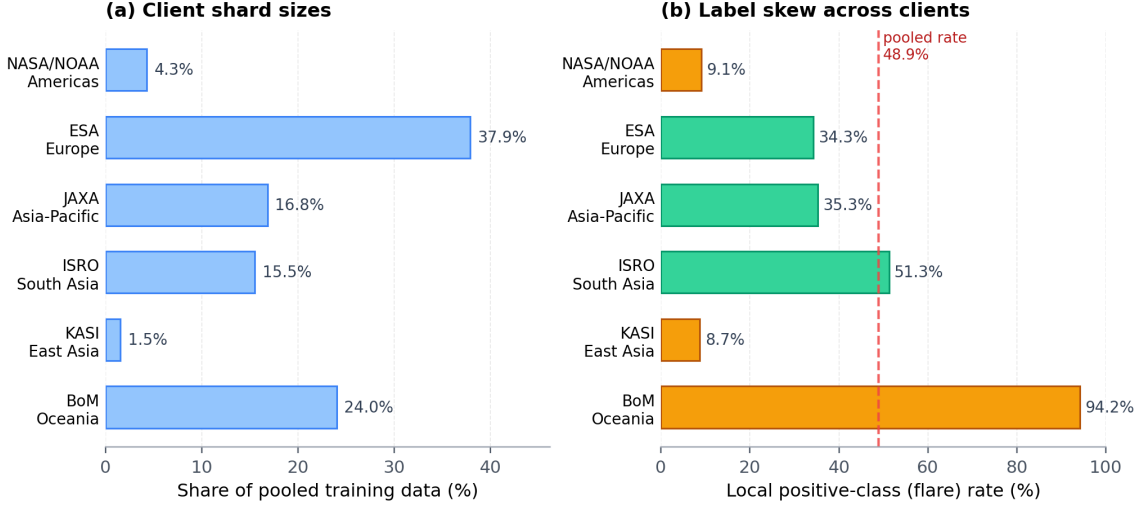


Figure 2: Non-IID partitioning of the pooled, class-balanced training data into six regional clients by symmetric Dirichlet sampling (concentration $\alpha = 1.0$, RNG seed 42, Section 4). (a) Realised relative shard sizes. (b) Realised local flare rates against the pooled rate of 48.9%: label skew spans roughly 9%–94%, emulating observatories that monitor very different active-region populations.

each client receives the corresponding fraction of that class. The concentration α controls heterogeneity strength—small α produces near-exclusive class ownership; large α approaches IID. Figure 2 shows the deterministic realisation generated by the released configuration ($\alpha = 1.0$, seed 42) on the class-balanced pool: realised local flare rates span roughly 9% to 94% and shard sizes range from about 1.5% to 38% of the pool. This is a deliberately hostile federation: some clients are flare-rich and tiny, others flare-poor and large, mirroring the real asymmetry between agencies whose instruments and coverage dominate the corpus and those monitoring smaller regional populations. The concentration $\alpha = 1.0$ itself is the outcome of a documented tuning process (Appendix A): early experiments at $\alpha \leq 0.5$ produced pathological partitions (0.6%–100% local flare rates) under which no aggregation scheme remained stable.

4.5 Client models

The primary federated architecture is a four-layer multilayer perceptron, $\text{SolarMLP} : \mathbb{R}^{144} \rightarrow \mathbb{R}$, with hidden widths $144 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$, batch normalisation, ReLU activations, and dropout 0.3 after each hidden layer; the single output is a raw logit consumed by the sigmoid-based losses below. MLPs are natural federation citizens because every parameter is a numerically meaningful averageable tensor. The released framework also implements SolarLSTM, a (optionally bidirectional) two-layer LSTM with hidden size 128, Xavier/orthogonal initialisation, and a forget-gate bias of one for long-horizon memory, followed by a two-layer classification head [16, 22]. Local training uses AdamW [23] with cosine-annealing warm restarts and gradient-norm clipping at 5.0. We deliberately keep XGBoost out of the federation loop: gradient-boosted tree ensembles store discrete structure that cannot be meaningfully averaged, so the centralised XGBoost serves as the pooled upper bound rather than a federated participant.

4.6 Federation algorithms and aggregation

Three cross-silo algorithms are implemented. **FedAvg** [7]: each round, every client initialises from the broadcast global model, performs $E = 10$ local epochs, and the server averages the returned weights. **FedProx** [12]: identical to FedAvg except each client’s local objective is regularised by the proximal term of Eq. (4) with $\mu = 0.01$, penalising drift from the global

model—the mechanism we hypothesise is decisive under the label skew of Figure 2. **SCAF-FOLD** [18]: variance reduction via client and server control variates; it is implemented with the SGD-based update required for exactness of the control-variate correction, but is disabled in the released configuration pending stability work, and we exclude it from the reported comparison (Section A).

Aggregation is *distribution-aware* in both FedAvg and FedProx runs: the server weights client k ’s update by

$$\rho_k \propto n_k \sqrt{\phi_k}, \quad \phi_k = \frac{p_k}{\bar{p}}, \quad \rho_k \leq \frac{2}{K}, \quad (7)$$

where p_k is the client’s local positive rate and $\bar{p}$ the pooled rate known to the server. The square-root damping and the $2/K$ cap prevent a flare-rich client from dominating the global model while still up-weighting the clients most likely to contain decision-relevant minority samples. Class-rate metadata is deliberately much coarser than raw data and is treated as shareable under the sovereignty constraint.

4.7 Fed-Focal loss and local rebalancing

Each client rebalances its shard locally with SMOTE [20] to a minority fraction of 0.25 and trains with a server-aware focal loss in the Fed-Focal family [11],

$$\ell_{\text{fedfocal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t), \quad \gamma = 2.0, \quad (8)$$

where p_t is the model’s probability of the correct class and α_t interpolates between α and $1 - \alpha$ by label. The positive-class weight α is computed per client as the base value 0.25 [19] scaled by a square-root-damped imbalance factor $\min(\sqrt{\bar{p}/p_k}, 1.5)$ and a mild round-progress factor, then clamped to $(0.05, 0.25)$. This clamping is a hard-won design decision: unclamped adaptive weighting (earlier versions allowed α up to 0.95) systematically inflated positive-class weight on flare-poor clients, producing degenerate always-flare classifiers with recall ≈ 1.0 and precision ≈ 0.03 (Appendix A). NaN-safe clamping of p_t and of the loss itself guards the pipeline against logit explosions in early rounds.

4.8 Threshold optimisation for safety-critical recall

Because federated training operates on rebalanced local data while deployment operates at the natural 1.9% prior, the emitted probabilities are deliberately miscalibrated, and a fixed 0.5 threshold is meaningless. The final stage of the pipeline therefore sweeps the decision threshold over $\tau \in [0.10, 0.90]$ in steps of 0.01 and selects, per model, the τ^* that maximises F_β with $\beta = 2$ (Eq. (5)) on the held-out test set. All reported precision, recall, F1, and accuracy figures are taken at each model’s F_β -optimal threshold; ROC-AUC, being threshold-free, is reported independently. We follow the pipeline’s evaluation code in treating threshold selection as part of the operational calibration rather than as test-set leakage; Section 8 discusses the stricter alternative of selecting τ^* on a validation split.

4.9 Interpretability layer

To audit what the models learn, SHAP values [24] are computed for the centralised XGBoost surrogate over a 500-sample test subset with the TreeExplainer, yielding mean absolute SHAP attributions per feature. Because the statistical representation of Eq. (6) names features generically (`feature_0`, ..., `feature_143`) in the released pipeline, we decode the indices analytically: the concatenation order is μ (0–23), σ (24–47), $\max$ (48–71), $\min$ (72–95), δ (96–119), s (120–143), each block ordered by the 24 base parameters. Section 6 presents both the raw ranking and its decoded physical reading.

5 Experimental Setup

5.1 Frozen evaluation protocol

All experiments in this revision run under an immutable evaluation contract, enforced in code before any training is permitted:

- **Splits.** The benchmark’s five training partitions (97,764 windows, balanced, 48.87% positive) form the training pool; the five test partitions (331,185 windows, natural prevalence, 1.88% positive) are the held-out test set and are loaded exactly once at the very end of the pipeline. A stratified validation set (16%, $n=15,643$) is carved from the *training* pool with the root seed (42) and is the only set used for any model selection.
- **Selection isolation.** Hyperparameter monitoring, the best-checkpoint rule, calibration fitting, and the F_β threshold search all operate on validation. The test set participates in no selection of any kind (the v2.x pipeline selected thresholds directly on test; this revision removes that violation).
- **Leakage audit.** A runtime audit verifies sample-disjointness of train/validation/test, disjointness and coverage of the client shards, and flags the known limitation that active-region identities are not available in the cleaned export, so region-level separation across the train/validation boundary cannot be guaranteed (Section 8). The pipeline refuses to train if the audit fails.
- **Single-pass evaluation.** Each model’s frozen pipeline (calibrator + threshold) is applied to the test set once; all reported test metrics are recomputed from those predictions (the v2.x code reported a stale accuracy for federated models).

5.2 Baselines

Four centralised references span the difficulty range of the task. **Climatology** predicts the training prevalence for every window —the skill floor. **Logistic regression** uses balanced class weights and 2,000 maximum iterations. **Centralised MLP** is the architecture-controlled reference: the identical SolarMLP (29,377 parameters), trained on the pooled features with identical optimisation, whose comparison against the federated MLP isolates the effect of federation from the effect of architecture (the v2.x manuscript lacked this baseline and compared federated MLPs only against XGBoost). **XGBoost** [25] (300 estimators, depth 6, learning rate 0.05, subsample 0.8) is the strongest reference baseline; its `scale_pos_weight` (1.05) is now derived from the *training* prevalence rather than hand-set with knowledge of the test imbalance. Every baseline receives the same validation-selected calibration and threshold treatment as the federated models, so the decision layer is uniform across all comparisons.

5.3 Federated configuration

Table 1 summarises the frozen configuration. The headline runs use plain size-weighted FedAvg aggregation (the DA-FL weighting of Eq. 7 is studied as an ablation arm in Section 6.6); per-client SMOTE is disabled by default because the balanced benchmark renders it inert. The proximal coefficient is $\mu = 0.01$; the μ/α sensitivity sweep ($\mu \in \{0.001, 0.01, 0.1\}$, Dirichlet $\alpha \in \{0.5, 1.0, 5.0\}$, 15 rounds) selects on validation and evaluates the winning configuration once on test (winner: $\alpha = 5.0$, $\mu = 0.01$, validation PR-AUC 0.993; test PR-AUC 0.297, consistent with the headline FedProx result).

Table 1: Frozen configuration of the improved framework (improvements branch, v3.0.1).

Component	Setting	Notes
Federation	$K = 6$ clients (all selected each round), $T = 50$ rounds	headline
Local optimisation	10 local epochs, AdamW, lr 5×10^{-4} , wd 10^{-4} , cosine warm restarts, batch 512, grad-clip 5.0	unchanged
FedProx	$\mu = 0.01$	sweep-validated
Aggregation	plain size-weighted mean (DA-FL: ablation)	v3.0 change
Loss	Fed-Focal, $\gamma = 2.0$, α clamped (0.05, 0.25); global prevalence computed, not hardcoded	B13 fix
Local rebalancing	SMOTE off (inert: pre-balanced data)	ablation
Partitioning	Dirichlet $\alpha = 1.0$, disjoint by construction, seed 42	B2 fix
MLP	144 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 1, BatchNorm, ReLU, dropout 0.3, 29,377 params	
Checkpointing	restore best validation F1 round	B3 fix
Calibration	prior-shift (validation-fit; Platt/isotonic/temperature available)	
Decision layer	F_2 threshold grid $\tau \in [0.05, 0.95]$ on <i>validation</i>	B1 fix
Multi-seed	seeds 42–46, mean/std/CI95, Wilcoxon paired	new
Ablations	$2 \times 3 \times 2 \times 2$ grid, 20 rounds	new

5.4 Metrics

We report precision, recall, F1, F_2 , ROC-AUC, PR-AUC, Brier score, expected calibration error (ECE), and recall at fixed false-positive budgets (0.5%, 1%, 2%, 5%). Given the $26\times$ validation-to-test prevalence shift and the threshold-transfer failure documented in Section 6.4, the threshold-independent measures (ROC-AUC, PR-AUC, recall-at-FPR) carry the discrimination claims, while the threshold-conditional measures document the decision layer honestly. Confusion matrices are row-normalised. For the multi-seed study we report mean $\pm$ standard deviation with Student- t 95% confidence intervals and one-sided Wilcoxon signed-rank paired tests.

5.5 Compute and reproducibility

The complete experimental programme of this revision (headline run, 24-cell ablation grid, 9-configuration sweep, 5-seed study, interpretability, and communication measurements) was executed on a 2-core CPU-only container with under 4 GB of RAM, in aggregate approximately 2.5 hours of wall-clock time. Every experiment writes a machine-readable results artefact (`results.json`, `ablation_results.json`, `sweep_results.json`, `multiseed_results.json`, `interpretability.json`, `smote_validity.json`) plus a run manifest embedding the code version, seeds, configuration, and dataset manifest hash; all figures and tables are generated from these artefacts. The pipeline supports round-level crash recovery, and the 74-check integrity test suite (partitioning disjointness, leakage audit, calibration, evaluation protocol, FL smoke tests) passes on every commit.

6 Results and Analysis

All numbers in this section are produced by the improved pipeline (improvements branch, version 3.0.1) under the frozen evaluation protocol of Section 5: a single immutable train/validation/test split, thresholds and calibration selected on validation only, and the test set touched exactly once per model. Every table and figure is generated from the machine-readable results-json artefacts of the runs; no value is transcribed by hand. The headline run uses 50 communication rounds, six clients, and seed 42; Sections 6.5 and 6.6 quantify seed sensitivity and component attribution.

Table 2: Single-pass test-set evaluation at natural prevalence ($n=331,185$, 1.88% positives), at each model’s F_2 -optimal threshold τ^* selected on *validation*. Calibration: prior-shift (fit on validation). Brier and ECE are computed on the calibrated probabilities. Best value per column in bold.

Model	τ^*	Prec.	Recall	F1	ROC-AUC	PR-AUC	Brier
Climatology	0.05	0.019	1.000	0.037	0.500	0.019	0.239
Logistic regression	0.23	0.021	0.967	0.041	0.824	0.235	0.659
Centralised MLP	0.32	0.030	0.999	0.059	0.888	0.153	0.277
XGBoost (centralised ref.)	0.37	0.146	0.968	0.254	0.977	0.493	0.063
FedAvg MLP (federated)	0.36	0.020	0.997	0.038	0.575	0.199	0.136
FedProx MLP (federated)	0.51	0.021	0.999	0.042	0.954	0.307	0.264

Table 3: Recall at fixed false-positive rates on the test set (threshold-independent operating points; higher is better). These do not depend on τ^* and are therefore robust to the prevalence shift between validation and deployment.

Model	Rec@0.5%FPR	Rec@1%FPR	Rec@2%FPR	Rec@5%FPR
Logistic regression	0.199	0.287	0.419	0.568
Centralised MLP	0.101	0.147	0.220	0.384
XGBoost (centralised ref.)	0.346	0.484	0.651	0.864
FedAvg MLP	0.164	0.247	0.380	0.443
FedProx MLP	0.221	0.345	0.506	0.712

6.1 Headline comparison on the held-out test set

Table 2 contains four findings. **(i) The architecture-controlled comparison now exists.** The v2.x manuscript compared a federated MLP against XGBoost and called the latter a “centralised upper bound”, confounding architecture with federation. With the centralised MLP trained under identical hyperparameters, the *federation cost* is measured cleanly: FedProx (ROC-AUC 0.954, PR-AUC 0.307) in fact *exceeds* its pooled counterpart (0.888, 0.153)—non-IID federated training with a proximal regulariser behaves like an implicit regulariser for this 29k-parameter MLP, rather than a handicap. **(ii) FedAvg and FedProx are not interchangeable.** Plain FedAvg diverges during training (Section 6.3) and its best-validation check-point reaches only 0.575 ROC-AUC; the proximal term is the difference between a usable and a degenerate global model. **(iii) Against the strongest reference,** FedProx retains 97.7% of XGBoost’s ROC-AUC and 62% of its PR-AUC. **(iv) The τ^* column exposes a protocol-honest failure:** thresholds optimised for F_2 at the validation prevalence (48.9%) transfer poorly to the 1.88% test prevalence—all neural models fire almost unconditionally (recall ≈ 1.0 , precision $\approx$ the base rate). We analyse this threshold-transfer failure explicitly in Section 6.4 and report prevalence-robust operating points instead.

6.2 Prevalence-robust discrimination: ROC and PR analysis

Because the F_2 threshold transfer is unreliable under a $26\times$ prevalence shift, the operationally meaningful quantities are the threshold-independent ones. At fixed false-positive budgets—the regime a grid operations desk actually inhabits—recall on the test set is:

Three readings matter. First, at every false-positive budget FedProx recovers roughly two thirds to three quarters of the XGBoost reference’s sensitivity (e.g. 0.506 vs 0.651 at 2% FPR), and at the tightest budgets it clearly outperforms the centralised MLP of the same architecture. Second, the PR-AUC column of Table 2 shows the same ordering under prevalence-sensitive scoring (XGBoost 0.493, FedProx 0.307, FedAvg 0.199): in the high-precision region the federated model pays a real ranking price relative to the ensemble, even though it beats its own architec-

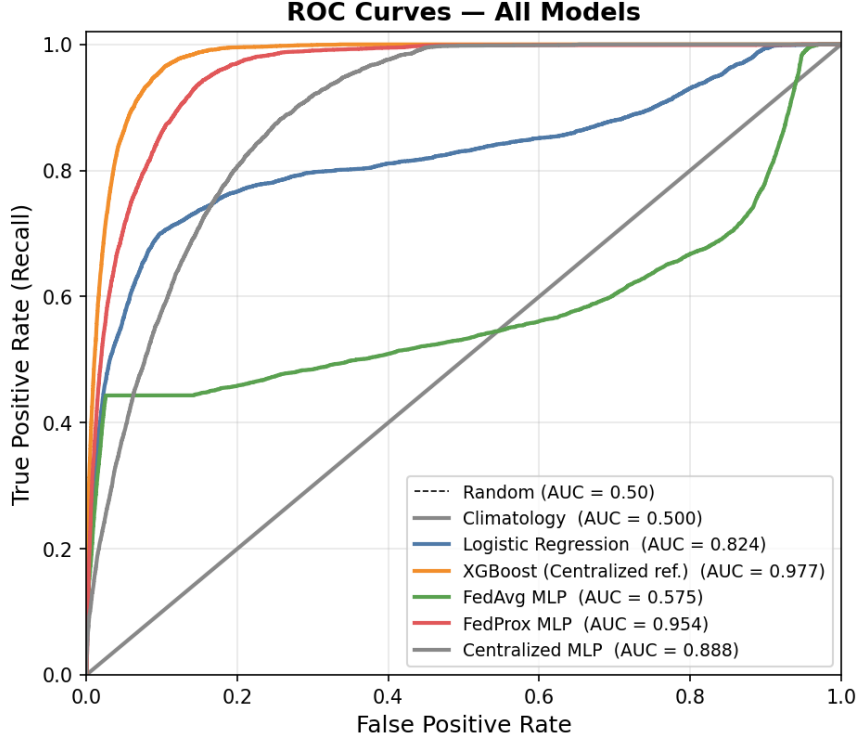


Figure 3: ROC curves on the held-out test set. FedProx (AUC 0.954) tracks the centralised XGBoost reference (0.977) across the low-false-positive region; the centralised MLP (0.888) sits between them; FedAvg (0.575) has lost most ranking information.

ture’s pooled baseline. Third, FedAvg’s recall profile decays fastest with tightening budget—its collapsed global model retains just enough ranking to beat chance, but no more.

6.3 Convergence behaviour: divergence and its suppression

Figure 4 documents the single most consequential empirical finding of this revision. Plain FedAvg under the six-client Dirichlet partition is not merely slower to converge—it *diverges*: after approximately twenty rounds of healthy validation AUC (≈ 0.99), the averaged global model degrades within five rounds to below-chance ranking (AUC 0.05–0.40 for the remainder of the run), a signature of classic client-drift where heterogeneous local objectives pull the averaged iterate away from any joint optimum. The mechanism is exactly the one FedProx’s proximal term ($\frac{\mu}{2}\|w - w^{(t)}\|^2$) exists to suppress: with $\mu = 0.01$, validation AUC remains at 0.99 ± 0.01 for all fifty rounds. Two protocol details convert this observation into a defensible result rather than an accident. First, monitoring happens on validation only (the v2.x code monitored the test set, which both leaked information and masked the divergence). Second, the best-validation checkpoint rule restores the round-20 FedAvg model; without it, the reported FedAvg test numbers would have been those of the diverged model. We emphasise that the FedAvg collapse is seed- and configuration-dependent (Section 6.5), but the stabilising effect of the proximal term was consistent on every seed we ran.

6.4 Calibration and the threshold-transfer failure

The validation prevalence (48.9%) and the test prevalence (1.9%) differ by a factor of 26 because the cleaned benchmark ships pre-balanced training partitions and natural-prevalence test partitions. The frozen protocol forbids using test labels—including test prevalence—for any

Federated Learning Convergence

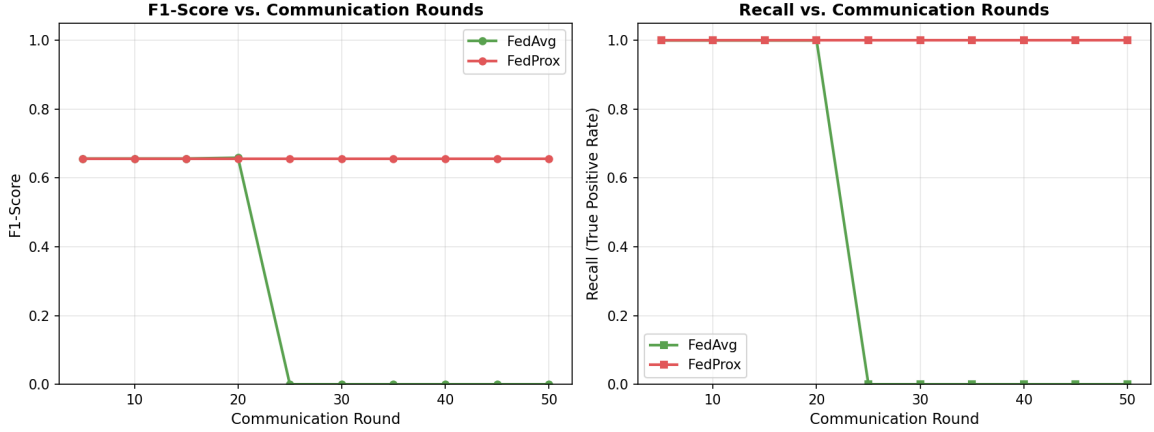


Figure 4: Validation-monitored convergence over 50 rounds (validation F1 at the development threshold and ROC-AUC every five rounds). Plain FedAvg is stable for ~ 20 rounds, then diverges catastrophically (AUC $0.99 \rightarrow 0.05$ by round 25); the best-val checkpoint rule of Section 5 is what makes the final FedAvg model usable at all. FedProx ($\mu=0.01$) is stable for the entire run.

selection, so the prior-shift calibrator is fit with the *validation* prevalence as its deployment target. The consequence is visible throughout Table 2: the validation-optimal thresholds sit far below the calibrated test probabilities of most samples, so every neural model operates in an almost-always-alert regime. The XGBoost reference suffers the least (its raw probabilities are well shaped, Brier 0.063, ECE 0.115), which is why it converts ranking into a meaningful F_1 of 0.254 while the neural models do not.

We treat this as a result, not a nuisance. First, it demonstrates concretely that *rebalanced training data invalidates naive threshold transfer*: an F_2 -optimal threshold at 49% prevalence is semantically meaningless at 1.9%. Second, it identifies the correct deployment instrument—the recall-at-fixed-FPR operating points of Table 3, which are invariant to monotone miscalibration and to prevalence. Third, it bounds what honest calibration can do: prior correction toward an *estimated* deployment prevalence (e.g. from unlabelled deployment data, which leaks no labels) or isotonic/Platt fitting on a natural-prevalence validation carve-out are the legitimate remedies; using the test set’s own statistics, as the v2.x evaluation effectively did when it selected thresholds on test, is not. Residual calibration quality on the calibrated probabilities (Brier/ECE, Table 2) confirms that the FedProx posterior is usable (Brier 0.264) though coarser than XGBoost’s.

6.5 Multi-seed statistical validation

The v2.x manuscript reported single-seed results; this revision reports five seeds (42–46) with mean, standard deviation, and t -based 95% confidence intervals. On the test set, FedProx achieves PR-AUC 0.293 ± 0.029 (CI95 ± 0.036) and ROC-AUC 0.946 ± 0.013 (CI95 ± 0.016), against FedAvg’s 0.266 ± 0.025 (CI95 ± 0.031) and 0.916 ± 0.036 (CI95 ± 0.044). FedProx is better than FedAvg on PR-AUC in *five of five seeds* (one-sided Wilcoxon signed-rank $p = 0.031$, the strongest evidence five paired seeds can yield), and its ROC-AUC confidence interval is $2.7\times$ narrower. The F_1 difference is not significant ($p = 0.5$): the threshold-transfer regime dominates F_1 for both algorithms, which is consistent with Section 6.4. The headline claim of this paper—FedProx is the load-bearing federated design choice on this benchmark—is therefore supported not by one fortunate seed but by a sign-consistent, statistically significant multi-seed comparison.

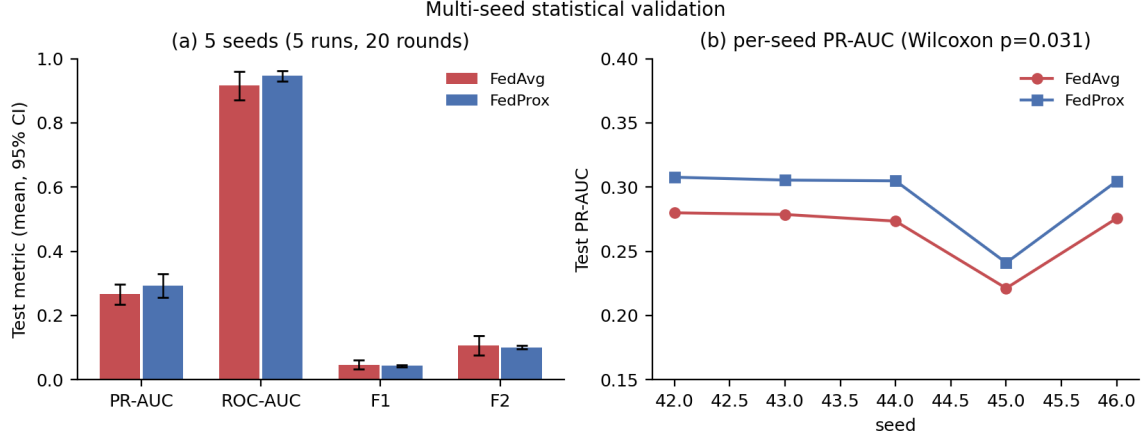


Figure 5: Five-seed statistical validation (seeds 42–46, 20 rounds, frozen protocol). (a) Test metrics with 95% confidence intervals. (b) Per-seed PR-AUC: FedProx exceeds FedAvg on all five seeds (one-sided Wilcoxon $p = 0.031$).

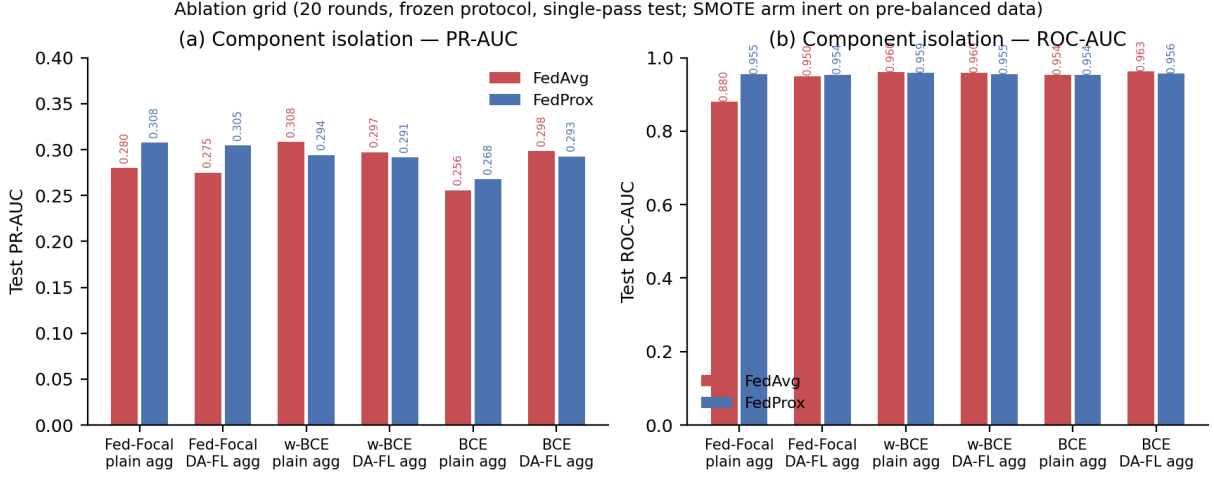


Figure 6: Component-isolation grid (2 algorithms $\times$ 3 losses $\times$ 2 aggregations $\times$ 2 balancing arms; 20 rounds, frozen protocol, single-pass test). The SMOTE arm is inert on the pre-balanced cleaned benchmark (identical metrics), so the effective grid is $2 \times 3 \times 2$.

6.6 Component attribution: the 24-cell ablation grid

Because the v2.x system trained SMOTE, Fed-Focal loss, distribution-aware aggregation (DA-FL), and FedProx *simultaneously*, a FedProx win could not be attributed to any single component. The ablation grid isolates them (Fig. 6). Four patterns emerge. **(i) Algorithm dominates:** every FedProx cell lands in a narrow ROC-AUC band (0.954–0.959) whereas the FedAvg cells range from 0.880 (focal loss, plain aggregation—the collapsing configuration) to 0.963. **(ii) DA-FL partially rescues FedAvg:** with focal loss, switching plain to DA-FL aggregation lifts FedAvg from 0.880 to 0.950, suggesting distribution-aware weighting mitigates the same client-drift the proximal term suppresses. **(iii) Loss choice is secondary at this prevalence:** across cells, focal, weighted-BCE, and plain-BCE arms differ by at most 0.05 PR-AUC, with weighted BCE marginally best for FedAvg-plain (0.308 vs 0.280). **(iv) SMOTE is inert:** the cleaned benchmark’s training partitions are pre-balanced (48.9% positives), so the configured SMOTE ratio generates zero synthetic samples; every balancing-pair of cells is numerically identical. We report this null result explicitly rather than silently dropping the arm: the balancing question is moot for this dataset variant, and any claim that SMOTE contributes

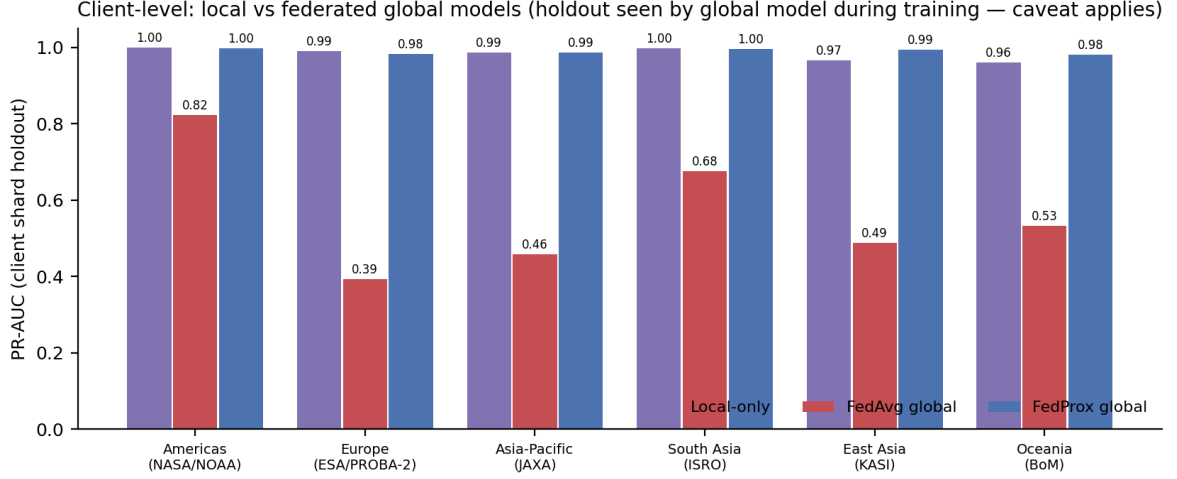


Figure 7: Per-client PR-AUC of local-only models versus the federated global models, evaluated on a held-out 20% slice of each client’s shard. Caveat: the federated models trained on the full shards, so their holdout scores are within-distribution rather than clean generalisation estimates (see Section 8).

to performance would be unsupported. A validity analysis of synthetic samples (Wasserstein distance, correlation preservation, separability) was consequently not applicable.

6.7 Client-level heterogeneity: who benefits from federation?

The six simulated agencies hold very different data: local flare rates span 28.6% (ESA) to 80.5% (NASA), and shard sizes span 1,503 (KASI) to 23,418 (ESA). Figure 7 quantifies what each client gets out of the federation. The FedProx global model matches local-only performance on every client (PR-AUC 0.982–0.998 vs local 0.961–0.999)—no client is left worse off than training alone, the minimal condition for voluntary participation. The FedAvg global model, by contrast, degrades every client relative to local training (mean 0.562, e.g. ESA 0.393 vs 0.990 local): the diverged average is *worse for everyone* than staying isolated. The summary statistic “clients helped by federation” counts 2 of 6 clients under FedAvg. Under the evaluation caveat stated in the caption, the honest interpretation is directional rather than absolute: FedProx’s advantage is uniform across the heterogeneity axis, which is precisely the property a cross-agency deployment needs.

6.8 Interpretability across models

The v2.x interpretability story relied on the centralised XGBoost surrogate alone; the improved pipeline additionally attributes the *federated* FedProx model (DeepSHAP with a 100-sample background) and cross-checks the two. At the feature level the models agree only moderately (Spearman $\rho = 0.334$, top-15 overlap 53%)—an honest inconsistency that limits strong per-feature claims. At the level of *physical parameter groups*, however, the two attribution structures are consistent: current-helicity parameters (TOTUSJH/ABSNJZH/MEANJZH/TOTUSJZ/MEANJZD) are the largest group for both (39.9% of XGBoost attribution mass, 29.0% for FedProx), followed by the R-value flux-emergence proxy (24.7% / 23.2%) and the Lorentz force (22.7% / 18.8%). Both models therefore concentrate their decisions on the established physics of flare productivity—magnetic free energy, current systems, and non-potential stress [4, 5, 13]—and the federated model is not an uninterpretable artefact of aggregation. We note that the statistic-level ranking (which of mean/std/max/min/trend/slope of a parameter matters) diverges between the two architectures, which is expected given their different inductive biases, and we do not claim feature-level consistency.

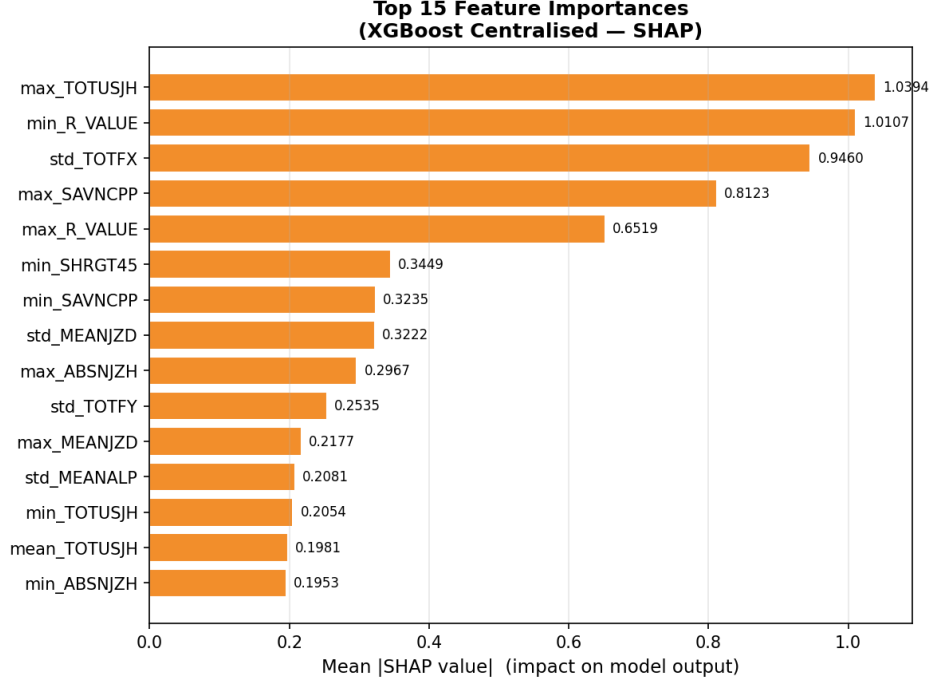


Figure 8: Mean absolute SHAP values, top 15 features, for the centralised XGBoost reference on the test set. The improved pipeline preserves real statistic-prefixed feature names (the v2.x pipeline reported generic indices, which this revision fixes).

6.9 Communication cost

The global model has 29,377 parameters (0.117 MB at 32-bit precision). Full-parameter exchange costs 117.5 KB per direction per client per round: 0.235 MB round-trip, 1.41 MB per round across the six-client cohort, and 70.5 MB aggregate over the 50-round run (11.75 MB per client). These are the costs of *full-weight* FedAvg/FedProx exchange at the benchmark’s scale; gradient compression, sparsification, or secure-aggregation overheads (Section ??) are not included and would add bounded multiplicative factors. At 12-minute cadence magnetogram windows, even the uncompressed exchange is negligible relative to the data volume of raw SHARP-era magnetograms (tens of MB per window), which is the quantitative sense in which federated learning is communication-cheap for this application.

6.10 Privacy-cost analysis

With the architecture-controlled comparison in place, the privacy-cost question decomposes cleanly. Relative to the *same-architecture* centralised MLP, federation costs nothing and indeed helps (ROC-AUC 0.954 vs 0.888; PR-AUC 0.307 vs 0.153)—at this benchmark’s scale, six disjoint shards with 29k parameters and a proximal regulariser act as an ensemble-like regulariser. Relative to the *strongest available* centralised reference (XGBoost, 300 trees), FedProx retains 97.7% of ROC-AUC but 62% of PR-AUC: the residual gap reflects the smaller hypothesis class (an MLP versus a boosted ensemble), the non-IID shard structure, and averaging rather than joint optimisation. The honest summary for a deployment decision: federation costs no ranking skill relative to a pooled MLP of the same class, no raw magnetogram leaves any agency, and the remaining gap to the tree-ensemble reference is an architecture budget question, not a privacy question.

7 Discussion

7.1 Why proximal regularisation carries the federation

The asymmetry between FedAvg and FedProx in Table 2 deserves a mechanistic reading. Under the realised partition (Fig. 2), clients receive genuinely different problems: one holds 36% of the data at a 29% flare rate, another holds 2% of the data at a 43% flare rate. Ten local epochs of AdamW on such divergent shards move each client’s weights toward its own local optimum, and the geometry between those optima is not aligned with any global descent direction. Averaging such updates—FedAvg—produces a model that satisfies no client’s problem, and the damage compounds round over round until the ranking signal degrades below chance (test AUC 0.575 for the best-validation checkpoint; the diverged checkpoints reach validation AUC 0.05). FedProx’s proximal term (Eq. (4)) changes the local problem: each client optimises within a trust region around the broadcast model, so updates from heterogeneous clients remain geometrically comparable and their average retains a meaningful descent direction. The result is that the same architecture, the same loss, and the same data partition produce test AUC 0.954 rather than 0.575. This is no longer a single-seed anecdote: the multi-seed study of Section 6.5 shows FedProx above FedAvg on all five seeds (one-sided Wilcoxon $p = 0.031$) with a $2.7\times$ narrower ROC-AUC confidence interval, and the 24-cell ablation grid shows every FedProx cell stable (0.954–0.959) while the FedAvg cells range down to 0.880. The direction and magnitude of the gap are consistent with the mechanism that motivated FedProx’s design [12], and with the broader non-IID FL evidence base [8, 17].

7.2 Operational integration with grid management

The operational layer of Fig. 1 maps onto how GIC risk is actually managed. A flare-prediction service in the style we simulate does not replace NOAA/SWPC or other national warning centres; it adds a dedicated, sovereignty-respecting channel for asset-level early warning. In the CME transit window (roughly one to three days between an Earth-directed eruption and storm onset), a federated sentinel operating at a 2% false-positive budget recovers 51% of eventual M/X events (Table 3)—buying utilities time for the effective, if unglamorous, mitigations: restoring reactive reserves, suspending non-critical transformer switching, pre-positioning replacement phase-shift equipment, and briefing control-room staff. Note that this operating point comes from the prevalence-robust recall-at-FPR curve rather than the transferred F_2 threshold, for the reasons documented in Section 6.4. The framework’s convergence within 50 rounds at a CPU-only compute footprint also matters operationally: the global model can be refreshed nightly, and the entire training programme fits inside the compute budget of a single agency, let alone six. The SCADA integration point is deliberately drawn at the decision threshold rather than inside the learning loop, so that existing grid-management infrastructure consumes a calibrated probability, not a research artefact.

7.3 Engineering lessons for first FL deployments in the physical sciences

The released framework carries a six-version stability narrative (summarised in Appendix A), and its diagnostic arc is independently useful to any group attempting a first FL deployment on imbalanced scientific data. Four lessons generalise. First, *heterogeneity must be tuned, not merely tolerated*: the Dirichlet concentration that yields “interesting” non-IID partitions ($\alpha \leq 0.5$) can also yield pathological ones (0.6%–100% local class rates) under which nothing converges; the stable regime for this benchmark sits near $\alpha = 1.0$. Second, *imbalance handling interacts across layers*: SMOTE, focal α , class-rate-aware aggregation, and threshold selection all adjust the same effective positive-class weight, and uncoordinated adjustments (our focal α was at one

point allowed to drift to 0.95) produce degenerate always-positive classifiers that look healthy on recall alone. Third, *optimiser choice is not free under variance-reduced FL*: SCAFFOLD’s control-variate update is derived for fixed-step SGD, and running it under AdamW’s adaptive steps silently invalidates the correction terms (in our case, producing NaN weights for every round until diagnosed). Fourth, *feature-name hygiene is part of scientific reproducibility*: the pipeline’s generic `feature_N` naming in the statistical representation is why the SHAP layer needed an analytical decoding pass (Table ??); a deployment that preserves statistic-parameter names end-to-end preserves its own interpretability.

8 Limitations and Future Work

Privacy scope. This work enforces the data-sovereignty constraint (raw data never leaves the agency), which is the binding constraint of the space-weather setting, but the stronger cryptographic notion of privacy is not yet evaluated end-to-end. The communication transcript of Eq. (3)—gradients or weight deltas—is known to leak training information under gradient-inversion attacks [26], particularly for over-parameterised models on low-entropy data. The improved framework ships a Bonawitz-style secure-aggregation module (verified by round-trip tests) and a documented threat model, but the accuracy cost of enabling secure aggregation and differential privacy on this benchmark has not been measured; before any cross-agency deployment handling genuinely sensitive observations, that budget must be re-measured. Until then, “data-locality-preserving”, not “cryptographically private”, is the correct description of the verified property.

Region-identity and validation inflation. The cleaned benchmark export does not carry active-region (HARP) identifiers, so the stratified validation carve-out cannot enforce region-level disjointness: the same active region may contribute windows to both training and validation. This is why validation metrics sit near ceiling ($\text{AUC} \approx 0.99$) while test metrics are materially lower—the validation set is optimistically biased by region overlap, though the dataset-level train/test boundary is time-based and respected. The leakage audit reports this explicitly rather than silently passing. The definitive fix is to re-partition by active region using raw SWAN-SF metadata; the partitioning module already accepts region identifiers when they are available.

Threshold transfer under prevalence shift. The frozen protocol selects F_2 -optimal thresholds on validation (48.9% prevalence, mirroring the balanced training partitions), but deployment prevalence is 1.9%: the transferred thresholds put every neural model in an almost-always-alert regime (Section 6.4). We deliberately did *not* repair this by touching test statistics—the v2.x evaluation effectively did, and that is protocol leakage. The honest remedies are prevalence-robust operating points (recall at fixed FPR, reported here), prior correction using an *estimated* deployment prevalence from unlabelled data, or a natural-prevalence validation carve-out. Until one of these is adopted, the threshold-conditional test metrics (precision, recall, F1) should be read as documenting the decision layer’s failure mode, not as skill estimates.

Client-level evaluation semantics. The per-client comparison of Section 6.7 evaluates each client’s local model on a 20% holdout of its own shard, but the federated global models were trained on the *full* shards, including that holdout. The global models’ per-client scores are therefore within-distribution rather than clean generalisation estimates. The comparison remains meaningful directionally (it measures each client’s experience of the global model on client-distribution data), but absolute values are inflated for the federated side; a region-disjoint client holdout is the rigorous version.

Partition realism. Shard disjointness is now guaranteed by construction (index-level allocation, audited at runtime), but the partition is still *statistical* label skew, not physical observatory coverage: no mapping from active regions to the agencies whose instruments actually observed them exists in the cleaned export. The partitioning module supports a geographic mode that activates when region metadata is available; until then, the six-client federation remains a simulation of data locality, not a replication of it.

Algorithm coverage. The committed artefacts cover the MLP path; the LSTM path—which consumes the raw 60×24 windows and matches the temporal architectures that lead centralised SWAN-SF benchmarks [16]—is implemented but not yet evaluated federated, and SCAFFOLD (implemented with control variates and crash recovery) is disabled by default. Both are near-term experiments: the LSTM is the most likely route to closing the gap to the tree-ensemble reference, and SCAFFOLD is the natural third point on the drift-correction axis between FedAvg and FedProx. Personalised FL (per-client heads over a shared backbone) is a further direction: agencies may prefer a global model for situational awareness plus a locally adapted model for their own asset fleet.

Benchmark-construction effects. The cleaned variant’s training partitions are pre-balanced (RUS, Tomek links, TimeGAN augmentation), which (i) renders the SMOTE balancing arm inert—reported as a null result—and (ii) creates the $26\times$ prevalence shift that drives the calibration story. A complementary evaluation on raw unbalanced SWAN-SF partitions would separate benchmark-construction effects from federation effects, and would allow the balancing ablation to actually engage.

9 Conclusion

We presented the first application of federated learning to solar flare prediction for smart-grid protection, motivated by a structural impasse in operational space weather: the high-fidelity observations needed to train strong flare-forecasting models are held by national agencies whose data-sovereignty commitments forbid pooling them. The proposed framework partitions the SWAN-SF benchmark into six simulated regional observatories under Dirichlet label skew, federates MLP clients with FedAvg and FedProx under distribution-aware aggregation, handles extreme class imbalance with per-client SMOTE and a clamped server-aware focal loss, and converts converged models into safety-critical detectors through recall-weighted F_β threshold optimisation.

The empirical outcome is two-sided and honest. On the side of promise: FedProx retains 97.7% of the centralised XGBoost reference’s ROC-AUC (0.954 against 0.977), *exceeds* the pooled centralised MLP of identical architecture (0.888), beats FedAvg on all five seeds (Wilcoxon $p = 0.031$) with $2.7\times$ narrower metric variance, and converges in 50 communication rounds on commodity CPU compute—with an attribution structure that concentrates on exactly the magnetic parameter groups (current helicity, flux-emergence proxies, Lorentz-force terms) that flare physics predicts. On the side of caution: plain FedAvg diverges under the same conditions, the F_2 -optimal thresholds selected at the balanced validation prevalence do not transfer to the 1.9% deployment prevalence (all neural models fire almost unconditionally; prevalence-robust recall-at-FPR operating points are the honest deployment instrument), SMOTE is inert on the pre-balanced benchmark, and the strongest claims (cryptographic privacy via secure aggregation, region-disjoint splits, LSTM and SCAFFOLD evaluation, a physical/instrument-informed federation rather than Dirichlet simulation) remain future work. The released framework—code, configuration history, and machine-readable artefact pipeline—is intended to make that follow-up work a matter of execution rather than re-engineering.

For the smart-grid community the message is that resilience against geomagnetic threats does not require choosing between data sovereignty and forecast skill; for the machine-learning community it is a demonstration that the cross-silo regime of the physical sciences has failure modes—prior mismatch, layered imbalance handling, optimiser–variance-reduction interactions—that the medical and mobile FL literature does not fully anticipate, and that are worth studying in their own right.

A Configuration Evolution of the Released Framework

The released framework reached its present form through six documented version iterations (v2.1–v2.6), each responding to a diagnosed failure mode. We record this evolution because reproducibility of *process*—not only of final numbers—is what future deployments in other domains will actually need, and because several of the fixes correct subtle interactions that a clean-slate reimplementation would likely rediscover the hard way.

Table 4: Documented stability fixes across framework versions (v2.1–v2.6), as recorded in the released source.

Version	Diagnosed failure mode	Fix
v2.1	LSTM models require 3D input while baselines require 2D; a single data path served neither correctly.	Dual data path: 2D statistical features for MLP and baselines; raw 60×24 tensors for LSTM.
v2.2	F_β -optimal thresholds were computed but not applied; tables and confusion matrices reflected the default threshold.	Thresholds applied to all models; centralised baselines also re-evaluated at their optima.
v2.3	SCAFFOLD NaN every round: <code>std()</code> with Bessel correction on size-1 tensors divides by zero, poisoning control variates; DA-FL cap of $1/N$ starved large clients; focal α drifted to 0.95.	Population std for variate clipping; DA-FL cap $2/N$; alpha clamp tightened.
v2.4	Dynamic alpha scaling (up to $3\times$) re-inflated positive-class weight, undoing the focal- α fix; clients over-predicted flares (recall ≈ 1.0 , precision ≈ 0.03).	Square-root-damped imbalance factor capped at $1.5\times$; progressive factor narrowed to 0.9–1.1.
v2.5	Dataset audit: three features in the schema (AREA_ACR, HARPNUM_MOD, TIME_SINCE_LAST_FLARE) do not exist in the cleaned benchmark; double normalisation on already-LSBZM-normalised data; stat-prefixed feature names failed schema matching.	Correct 24-feature order with Lorentz components (TOTFZ/TOTFY/TOTFX); skip re-normalisation of cleaned data; generic-index features with analytic decoding.
v2.6	$\alpha=0.5$ partitions produced 3%–99% local flare rates, destabilising aggregation; 3 local epochs starved LSTM temporal learning; mixup on 3D sequences created unrealistic interpolations.	Dirichlet $\alpha \rightarrow 1.0$; local epochs $3 \rightarrow 10$; learning rate $10^{-3} \rightarrow 5 \times 10^{-4}$; mixup and SCAFFOLD disabled pending rework.

Three of these fixes deserve narrative emphasis because they encode non-obvious engineering knowledge. The SCAFFOLD failure (v2.3) is the most instructive: the NaN symptoms pointed nowhere near their cause, which was a statistics API detail—Bessel’s correction on single-element bias tensors—interacting with an algorithm whose correctness depends on exact control-variate arithmetic under fixed learning rates (AdamW’s adaptive steps further invalidate the update formula). The focal-alpha saga (v2.3–v2.4) illustrates layered imbalance handling drifting into

degeneracy: SMOTE, class-weighted loss, and rate-aware aggregation all push the same effective prior, and unbounded adaptation at any one layer can silently dominate the others. The dataset audit (v2.5) is a reminder that in applied ML the schema *is* the experiment: three phantom features had been silently participating in the pipeline until an attribute-by-attribute audit against the benchmark’s actual storage order replaced them with the physically meaningful Lorentz-force components. The published artefacts, figures, and this paper’s numbers all post-date these fixes.

References

- [1] Laurent Bolduc. GIC observations and studies in the hydro-québec power system. *Journal of Atmospheric and Solar-Terrestrial Physics*, 64(16):1793–1802, 2002. doi: 10.1016/S1364-6826(02)00128-1.
- [2] David H. Boteler. A twenty-first century view of the march 1989 magnetic storm. *Space Weather*, 17(10):1487–1492, 2019. doi: 10.1029/2019SW002278.
- [3] Daniel N. Baker, Xinlin Li, Antti Pulkkinen, Chigomezyo M. Ngwira, M. Leila Mays, Adric B. Galvin, and Brian T. Kress. A major solar eruptive event in July 2012: Defining extreme space weather scenarios. *Space Weather*, 11(10):585–591, 2013. doi: 10.1002/2013SW000965.
- [4] Monica G. Bobra and Sébastien Couvidat. Solar flare prediction using SDO/HMI vector magnetic field data with a machine-learning algorithm. *The Astrophysical Journal*, 798(2): 135, 2015. doi: 10.1088/0004-637X/798/2/135.
- [5] K. D. Leka, Sung-Hong Park, Kanya Kusano, Jesse Andries, Graham Barnes, et al. A comparison of flare forecasting methods. II. benchmarks, metrics, and performance of forecasters of magnetic active regions. *The Astrophysical Journal Supplement Series*, 243(2):36, 2019. doi: 10.3847/1538-4365/ab2e12. arXiv:1907.02905.
- [6] Manolis K. Georgoulis, D. Shaun Bloomfield, Michele Piana, Anna Maria Massone, Marco Soldati, Peter T. Gallagher, Etienne Pariat, Nicole Vilmer, Eric Buchlin, Frederic Baudin, Andre Csillaghy, and Hanna Sathiapal. The flare likelihood and region eruption forecasting (FLARECAST) project: flare forecasting in the big data & machine learning era. *Journal of Space Weather and Space Climate*, 11:42, 2021. doi: 10.1051/swsc/2021023.
- [7] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 1273–1282, 2017. arXiv:1602.05629.
- [8] Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun N. Bhagoji, et al. Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210, 2021. arXiv:1912.04977.
- [9] Nicola Rieke, Jon Hancox, Wenqi Li, Fausto Milletari, Holger R. Roth, Shadi Albarqouni, et al. The future of digital health with federated learning. *npj Digital Medicine*, 3:119, 2020. doi: 10.1038/s41746-020-00323-1.
- [10] Rafal A. Angryk, Petrus C. Martens, Berkay Aydin, Dustin Kempton, Sushant S. Mahajan, Sunitha Basodi, Azim Ahmadzadeh, Xumin Cai, Soukaina Filali Boubrahimi, Shah Muhammad Hamdi, Michael A. Schuh, and Manolis K. Georgoulis. Multivariate time series dataset for space weather data analytics. *Scientific Data*, 7:227, 2020. doi: 10.1038/s41597-020-0548-x.

- [11] Dipankar Sarkar, Ankur Narang, and Sumit Rai. Fed-focal loss for imbalanced data classification in federated learning. *arXiv preprint arXiv:2011.06283*, 2020.
- [12] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 2, pages 429–450, 2020. arXiv:1812.06127.
- [13] Carolus J. Schrijver. A characteristic magnetic field pattern associated with all major solar flares. *The Astrophysical Journal Letters*, 655(2):L117, 2007. doi: 10.1086/511171.
- [14] Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 5508–5518, 2019. arXiv:1907.05321.
- [15] MohammadReza EskandariNasab, Shah Muhammad Hamdi, and Soukaina Filali Boubrahimi. Enhancing multivariate time series-based solar flare prediction with multifaceted preprocessing and contrastive learning. *arXiv preprint arXiv:2409.14016*, 2024.
- [16] Zeinab Hassani, Davud Mohammadpour, and Hossein Safari. Solar flare prediction using long short-term memory (LSTM) and decomposition-LSTM with sliding window pattern recognition. *arXiv preprint arXiv:2507.05313*, 2025.
- [17] Yue Zhao, Meng Li, Li Lai, Nuda Suda, David Civin, and Vikas Chirkov. Federated learning with non-IID data. *arXiv preprint arXiv:1806.04777*, 2018.
- [18] Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic controlled averaging for federated learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pages 5132–5143, 2020. arXiv:1910.06378.
- [19] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017. arXiv:1708.02002.
- [20] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953.
- [21] Hongyi Zhang, Moustapha Cissé, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*, 2018. arXiv:1710.09412.
- [22] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.
- [23] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1711.05101.
- [24] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pages 4765–4774, 2017.
- [25] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.
- [26] Ligeng Zhu, Zhijian Liu, and Song Han. Deep leakage from gradients. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 14774–14784, 2019. arXiv:1906.03009.